

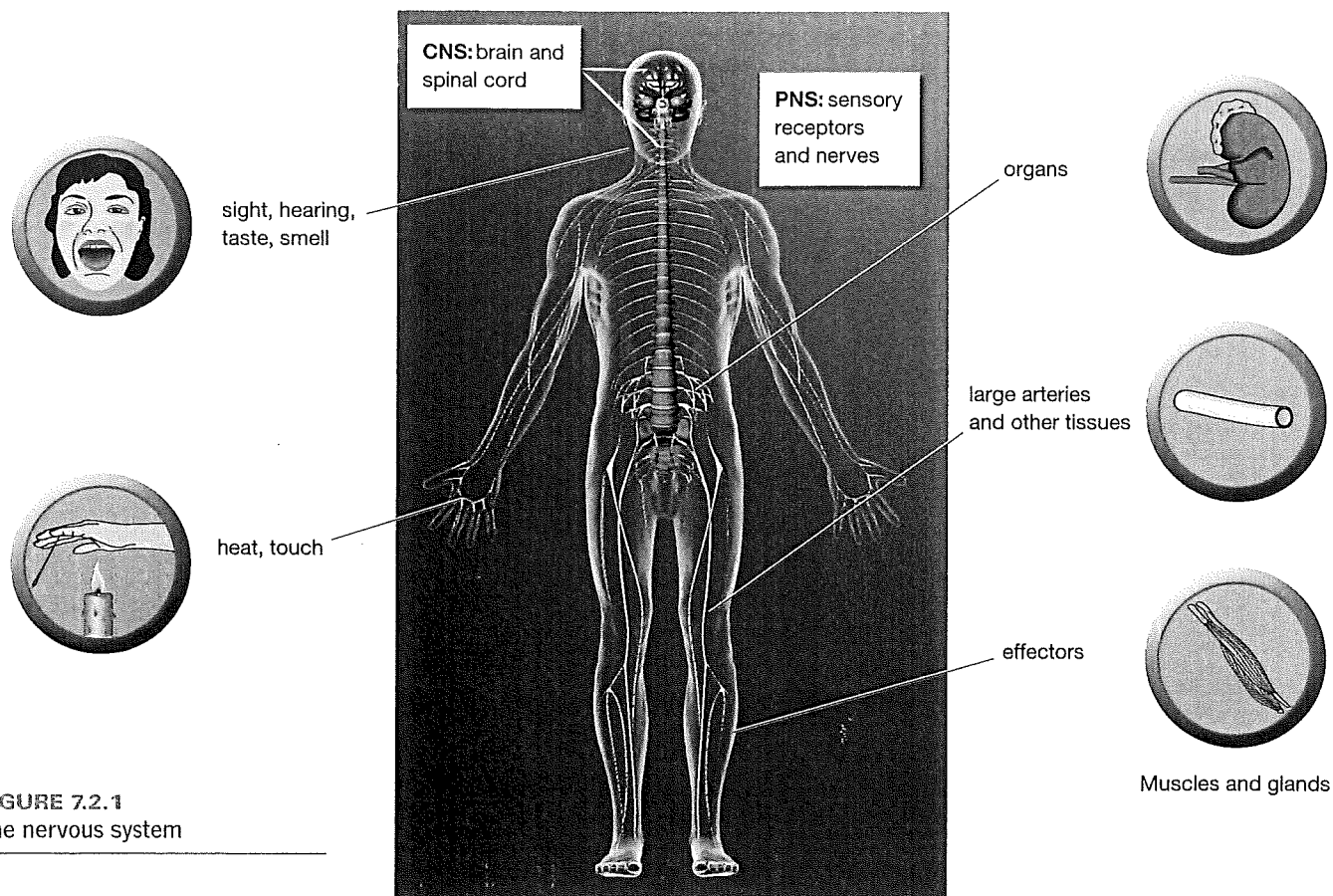


FIGURE 7.2.1  
The nervous system

## Nerve cells

The nervous system is made up of trillions of nerve cells called **neurons**. **Neurons** are specialised cells that transmit electrical messages from one part of your body to another at very high speed. These electrical messages are called **nerve impulses** and they can travel in only one direction.

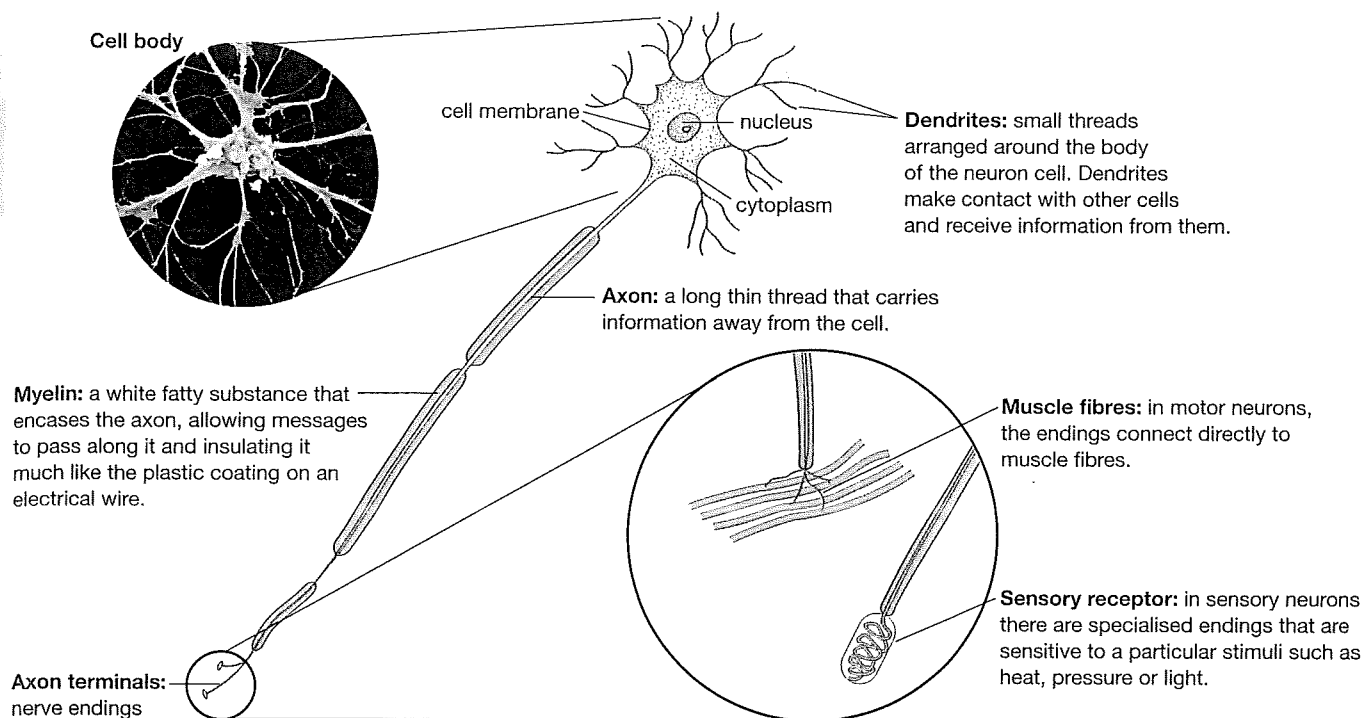
A neuron has all of the usual features of any animal cell including a nucleus, cell membrane and cytoplasm. As you can see in Figure 7.2.2, a neuron also has many other specialised parts including dendrites, an axon and axon terminals.

The **cell body** contains the nucleus, which is the control centre of the cell. The **dendrites** branch out from the cell body and receive messages from other nerve cells, which are then sent on to the cell body. The **axon** or nerve fibre sends nerve impulses in only one direction—away from the cell body. The axon terminals pass the message on to the next neuron.

There are different types of neurons. Each type of neuron has a different function within the nervous system (Figure 7.2.3).

- **motor neurons**—carry messages from the CNS to effectors. **Effectors** are muscles or glands (tissues that secrete hormones) that translate the messages into actions.
- **connector neurons**—transmit messages between neurons in the CNS.
- **sensory neurons**—have specialised receptors, which are sensitive to stimuli such as heat or light. They carry messages to the brain and spinal cord from cells in the sense organs (such as your eyes, ears, tongue and skin).

The messages sent along the neuron are electrical. If all the neurons in your body touched one another, then stimulating one nerve ending would be like turning on one switch in your house and having all the lights and appliances come on. Your body needs to control which nerves 'fire' at a certain time.



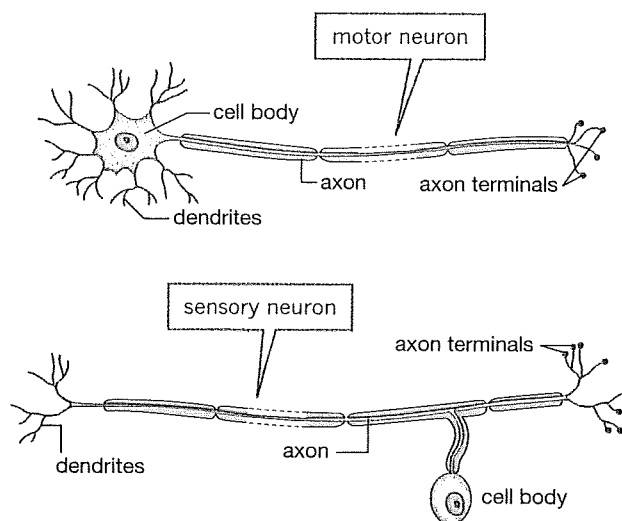
**FIGURE 7.2.2** These are some of the specialised endings a typical neuron can have.

When the nerve impulse reaches the axon terminals at the end of an axon, a chemical called a **neurotransmitter** is released into the space between the neurons. This space is called a **synapse** (Figure 7.2.4 on page 282). The neurotransmitter carries the message from the axon of one neuron to the dendrite of the next neuron. The dendrite receives the chemical message and sends off an electrical signal.

About 50 different neurotransmitters carry electrical impulses across these gaps. These neurotransmitters control which nerves fire and when.

In your body, the neurons are bundled together to form nerves, as shown in Figure 7.2.5 on page 282. Neurons are covered with an insulating layer called a **myelin sheath**. The myelin sheath electrically insulates the neurons from each other and increases the speed of the nerve impulse.

The parts of the CNS that contain neurons covered in myelin are called white matter. The parts that contain mainly cell bodies are called grey matter. The outer parts of the brain are made up mainly of grey matter.

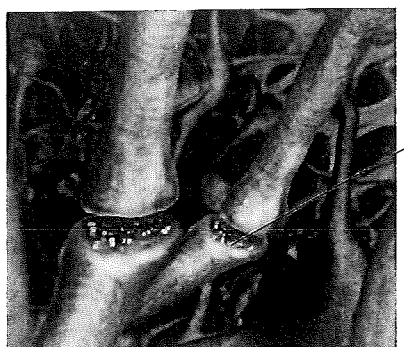


**FIGURE 7.2.3** Two types of neuron

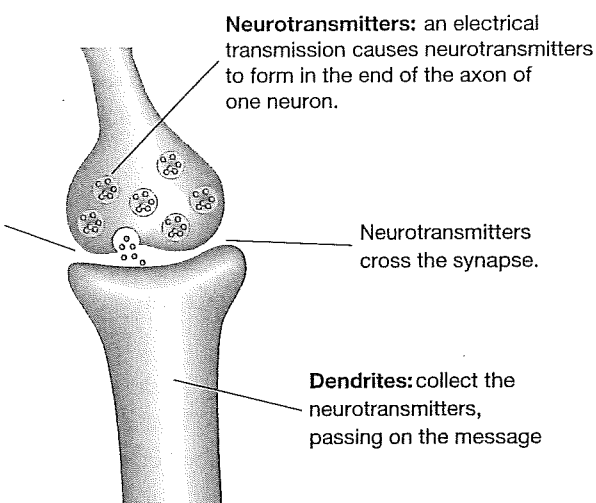
## SciFile

### What a nerve!

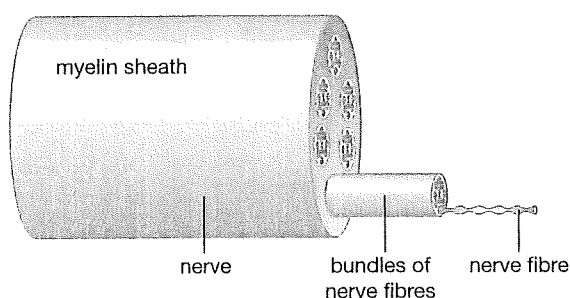
The longest neuron in your body extends from your big toe to the middle part of your spine and is about a metre long. A giraffe's longest nerve is in its neck and is up to 5 metres long!



**Synapse:** a small gap between neurons. Slows the message and allows it to be redirected to different neurons and parts of the body.



**FIGURE 7.2.4** At the synapse, the electrical signal of the nerve is converted into a chemical signal called a neurotransmitter, which carries the signal across the gap. The chemical signal is then converted back to an electrical signal.



**FIGURE 7.2.5** A nerve is made up of a large number of neurons, each of which is surrounded by a myelin sheath.

AB  
7.3

## The brain

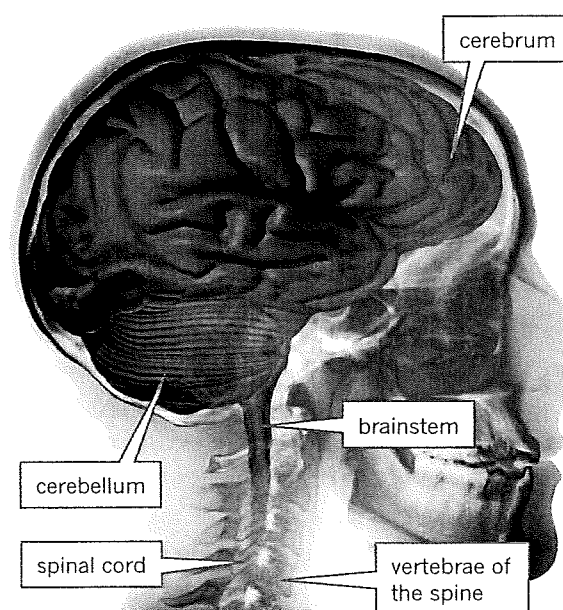
Compared with other animals, humans have a very large brain for their body size. The human brain contains about 100 billion neurons, and has an average volume of 1200–1400 mL. The brain controls and regulates body functions. Without it, you cannot survive.

Medical imaging techniques can look inside a living brain. For example, MRI (magnetic resonance imaging) uses strong magnetic fields to distinguish different types of body tissue. MRI is useful in diagnosing brain tumours and finding areas of brain injury.

Sometimes other parts of the brain take over the function of the damaged parts, but there are situations where brain damage is permanent. fMRI (functional magnetic resonance imaging) measures and maps brain activity through changes in blood flow. fMRI is useful to determine the effects of a stroke or disease, or to guide brain treatment.

## The cerebrum

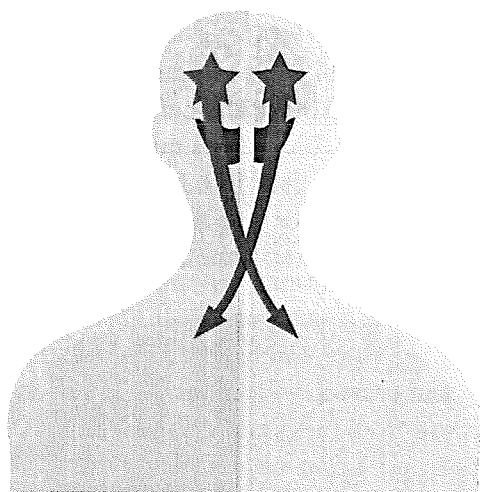
When you think of what a human brain looks like, you are probably thinking of the **cerebrum**. You can see it in Figure 7.2.6. It occupies more than 80% of the brain and contains over 10 billion neurons. Its folded surface makes surface area three times greater than if the brain's surface was smooth.



**FIGURE 7.2.6** Computer-enhanced X-ray showing the external structure of a human brain.

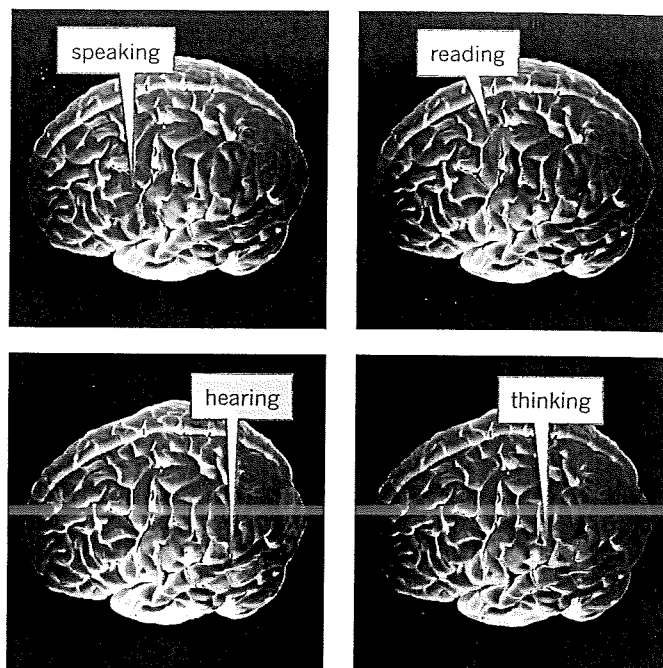
It is in the cerebrum that the higher intellectual functions of humans take place. The cerebrum controls your conscious thoughts and the intentional (voluntary) movement of every body part. For example, if you scratch your nose then it is the cerebrum controlling your movement. The cerebrum also receives sensory messages from all body parts. For example, physical pain, the sound you hear and the light you see are all processed by the cerebrum.

The cerebrum is made up of two parts, called the right and left cerebral hemispheres. When you are performing an intended and voluntary action such as walking or hitting a ball, the right hemisphere controls the left side of your body and the left hemisphere controls the right side of your body (Figure 7.2.7). Each half of the brain can work independently, but you use both cerebral hemispheres for most activities. One side usually dominates in a particular task. For example, in most people the left side has more control over language and logical thinking, such as mathematical ability. The right side is the more creative and emotional side. Musical and artistic abilities depend on the right side of the brain.



**FIGURE 7.2.7** The right and left sides of the brain control the opposite sides of the body.

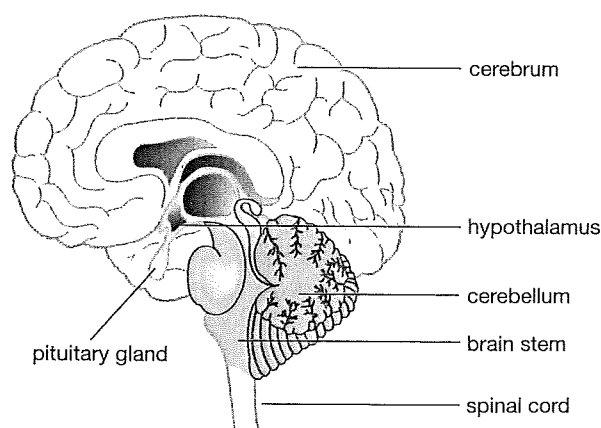
Figure 7.2.8 shows images of the brain created using both fMRI and PET (positron emission topography). Together these scans reveal the parts of the brain that are active during various activities. They show that the left-hand side of the brain is active during activities that involve language. They also show that different parts are active when listening, speaking, reading or just thinking about words.



**FIGURE 7.2.8** The red and green areas in these images show the areas of the brain that are active during various activities.

## The cerebellum

At the base of the cerebrum is the **cerebellum**. The cerebellum is located where your skull curves inwards (Figure 7.2.9). The cerebellum is responsible for coordination and balance. Without it, walking would be impossible.



**FIGURE 7.2.9** Vertical cross-section of the brain

Prac 1  
p. 290

Prac 2  
p. 291

Prac 3  
p. 292

## The medulla and spinal cord

The lower part of the **brain stem** or **medulla** can be seen where the spinal cord widens just after it passes into the skull. It controls the body's vital functions, such as breathing, blood pressure and heart rate. Damage to this area can be fatal.

## Protecting the brain and spinal cord

The spinal cord and brain are so important to body function that they have special protective structures surrounding them. A covering of bone known as the **cranium** protects the brain and the spinal cord is also surrounded in bony structures called **vertebrae**. Both the spinal cord and the brain are surrounded in a fluid called **cerebrospinal fluid (CSF)**, which provides nutrients to the neurons and acts as a shock absorber.

## science 4 fun

### Stroop effect

How easy is it to fool your brain?

Do this ...

As quickly as you can, name the colour the word is printed in. Do NOT read the actual words.

RED

BLUE

GREEN

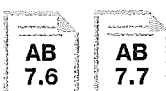
ORANGE

YELLOW

VIOLET

Record this ...

- 1 Describe what happened.
- 2 Explain why you think this happened.



## Responding to stimuli

A simple model of your nervous system is a stimulus-response model. You can see it in Figure 7.2.10. Receptors stimulate the sensory neurons. This is referred to as the **stimulus**. The sensory neurons send a message to the brain. The brain works out the response that is required, then sends a message along motor neurons to the effectors.

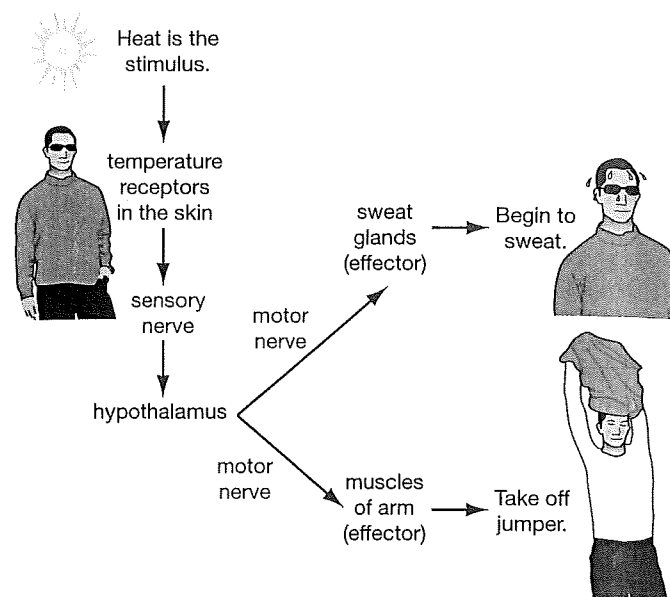


FIGURE 7.2.10 This stimulus-response outlines your body's reaction to being too hot.

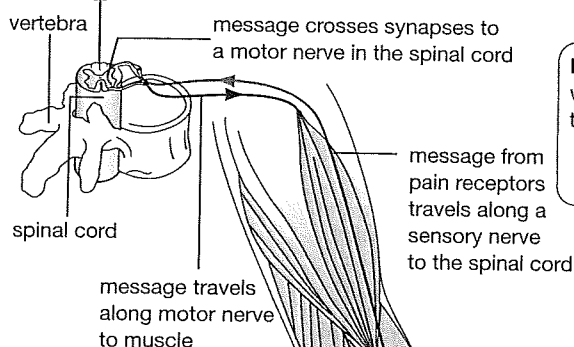
## Reflex actions: a rapid response

Arm muscles are normally under voluntary control. This means that you normally control what they do. However, when you touch something hot or sharp, you automatically pull your hand away. This reaction is involuntary and very fast—your arm didn't need to wait for instructions from the brain. This reaction is called a **reflex action** and it protects your body from danger.

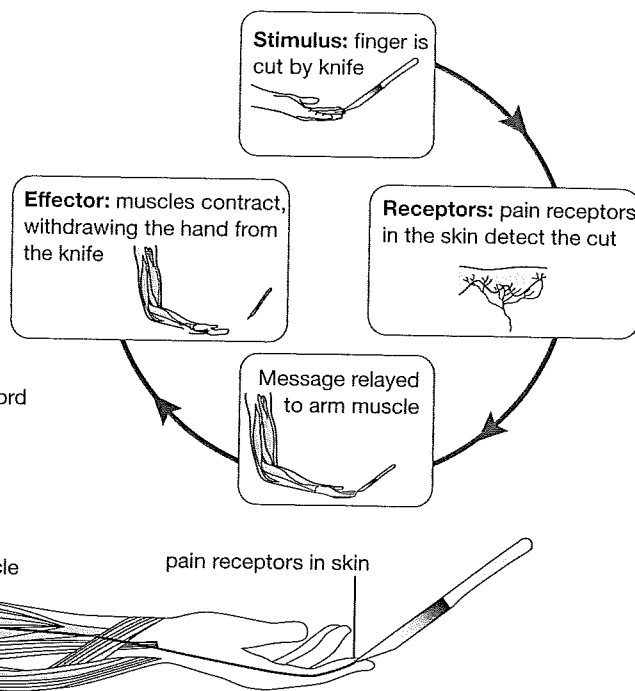
Receptors detect the heat on your skin. This heat activates a sensory neuron, which sends nerve impulses to the spinal cord. Within the spinal cord, a relay neuron passes the message directly to a motor neuron, which sends impulses to the arm muscles, which are the effectors. The arm muscles contract, lifting your hand away from the sharp object. A message is sent to the brain shortly afterwards.

Only then can the brain register pain. The nerve pathway operating in a reflex action is called a **reflex arc**. Figure 7.2.11 shows an example of a reflex arc. Most reflex actions are rapid because they only involve a few neurons.

a message may be sent to the brain but only to keep it informed of what is happening. The brain might trigger tears or a cry of pain



**FIGURE 7.2.11** A reflex arc does not require conscious thought as it does not directly involve the brain.



AB  
7.4

## STEM 4 fun

### Model arm

#### PROBLEM

Can you build a model arm?

#### SUPPLIES

- 2 icy-pole sticks (or cardboard strips of a similar size)
- box knife
- wire
- cotton or string
- scissors
- stationery split pin
- skewer etc.

**SAFETY**

Take extreme care when using the box knife.

**PLAN AND DESIGN** Your task is to build a model arm with icy-pole sticks for the bones and string for the muscles that act as effectors. Use it to show that muscles make the bones move.

Design the solution. What information do you need to solve the problem? Draw a diagram. Make a list of materials you will need and steps you will take.



**CREATE** Follow your plan. Produce your solution to the problem.

**IMPROVE** What works? What doesn't? How do you know it solves the problem? What could work better? Modify your design to make it better. Test it out.

#### REFLECTION

- 1 What field of science did you work in? Are there other fields where this activity applies?
- 2 In what career do these activities connect?
- 3 What did you do today that worked well? What didn't work well?

## Autonomic nervous system

The word autonomic means 'independent'. Consequently it is an appropriate name for the part of the nervous system which controls the unconscious internal affairs of the body; that is, those processes which are largely independent of thought.

For example, when a man runs to catch a bus his muscles are controlled by the conscious parts of his nervous system. But, at the same time, his autonomic nervous system automatically increases the rate of his breathing and heart-beat, dilates the blood vessels which carry blood to his muscles, releases sugar from his liver, and makes many other adjustments to provide for the sudden burst of activity. When the man reaches the bus and sits down his autonomic nervous system slows down all the processes which it previously speeded up. To summarize, the autonomic system regulates the body's internal environment, making automatic adjustments for activity and rest.

The autonomic nervous system consists of two parts: the **sympathetic** and the **parasympathetic**. These two systems are **antagonistic** to one another, which means they have opposite functions. The sympathetic system is mostly concerned with preparing the body for action while the parasympathetic system prepares the body for rest.

The sympathetic and parasympathetic systems are able to have opposite effects on the body because most organs and glands are penetrated by nerves from *both* systems (Fig. 1). This is called **double innervation**. When the iris of the eye, for example, receives impulses from its sympathetic nerve the pupil is dilated (enlarged), whereas parasympathetic stimulation contracts the pupil.

### How sympathetic nerves prepare the body for action

In an emergency the sympathetic nervous system is able to have immediate widespread effects on the body for two reasons. First, after each sympathetic nerve fibre leaves the central nervous system it links up with twenty to thirty other fibres which transmit impulses simultaneously to many different organs and glands. Second, sympathetic nerves stimulate the adrenal glands to produce the hormone **adrenalin**. This hormone (described in Unit 54) is distributed quickly by the blood and stimulates organs into greater activity.

Sympathetic and adrenalin stimulation have the following effects on the body.

1. The heart-beat is accelerated and coronary blood vessels dilate, increasing blood supply to heart muscle.

2. Muscle in the bronchiole walls relaxes. This increases the diameter of these tubes so that the lungs can be inflated and deflated at greater speed.

3. Blood vessels in the skin and gut contract, reducing blood supply to these regions and diverting it to the limb muscles whose blood vessels are dilated.

4. In the liver, glycogen is converted into glucose which pours into the blood and around the body.

When the emergency is over the parasympathetic system restrains the heart-beat, contracts the bronchioles, increases blood supply to the skin and gut, and switches the liver over to converting excess blood glucose into glycogen. These and other actions adjust unconscious processes to a level suitable for leisurely conscious activity, thereby avoiding waste and excessive wear and tear on organs.

Table 3 Effects of autonomic stimulation

| <i>Sympathetic stimulation</i>                     | <i>Parasympathetic stimulation</i>                  |
|----------------------------------------------------|-----------------------------------------------------|
| Dilates pupil of eye                               | Constricts pupil                                    |
| Increases heart rate                               | Decreases heart rate                                |
| Dilates coronary blood vessels                     | Constricts coronary vessels                         |
| Dilates bronchi and bronchioles                    | Constricts bronchi and bronchioles                  |
| Reduces peristalsis and closes gut sphincters      | Increases peristalsis and opens gut sphincters      |
| Nil                                                | Causes production of digestive juices               |
| Relaxes bladder wall and closes bladder sphincters | Contracts bladder wall and opens bladder sphincters |
| Converts glycogen to glucose in the liver          | Converts glucose to glycogen in the liver           |
| Causes production of sweat                         | Nil                                                 |



## Senses in the skin, tongue, and nose

### Skin

There are at least five different types of sensory nerve ending in the skin. These, and all other sensory nerve endings, are called **receptors**, because they are the parts of the nervous system at which information is received. It is not yet certain how skin receptors work or what their individual functions are. Nevertheless they have been named according to their most likely functions, which are sensitivity to: touch, pressure, pain, temperature, and hair movements (Fig. 1).

**Touch** These receptors are located immediately below the epidermis, and are most numerous in the surface of the tongue and fingertips. Touch receptors are stimulated by light pressure on the skin and enable a person to distinguish between textures such as rough, smooth, hard, and soft. A different type of touch receptor detects hair movements caused by wind or objects brushing against the skin.

**Pressure** These receptors are situated beneath the dermis and are stimulated by heavy pressure.

**Pain** These receptors consist of branched nerve endings in the epidermis and dermis. They are quite evenly distributed throughout the skin and occur almost everywhere inside the body except the brain. Pain is unpleasant but very important because it is a warning signal that something is going wrong inside the body, or that some part of the skin is being damaged.

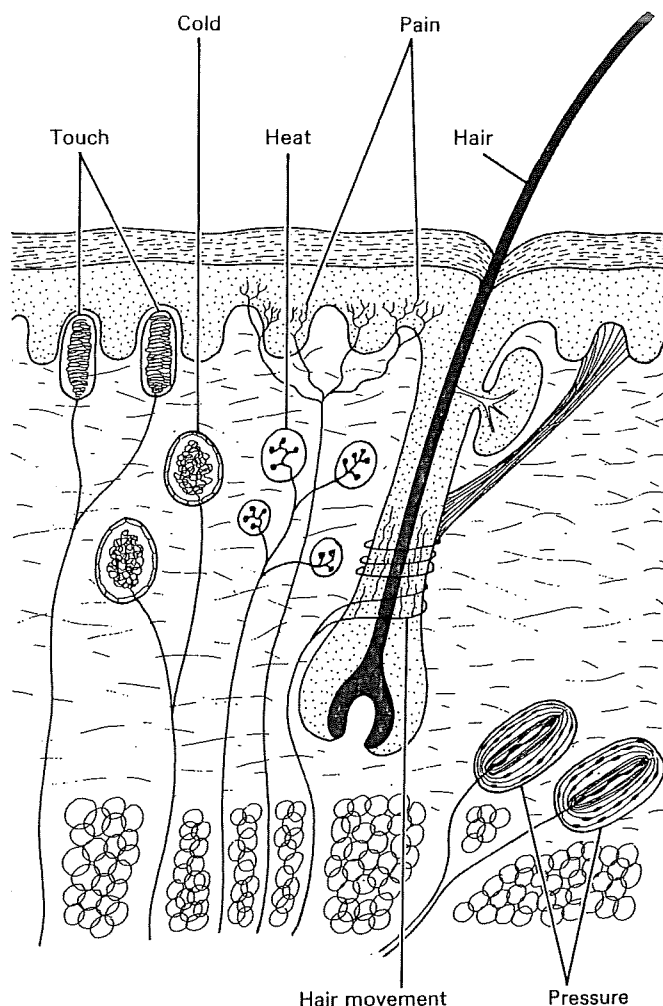
**Temperature** There appear to be separate 'heat' and 'cold' receptors in the skin. They are stimulated by sudden changes in temperature.

### Nose

Smell receptors, or olfactory organs, are sensitive to chemicals in the air (Fig. 2). These chemicals must first dissolve in a film of moisture which covers the receptors.

### Tongue

Taste receptors, or taste buds, in the tongue are also sensitive to chemicals (Fig. 3B). All taste buds look alike (Fig. 3C) but in fact there are four different types: those which respond to salt, sweet, sour, and bitter-tasting substances. Groups of each type of receptor are concentrated in certain areas of the tongue (Fig. 3A). The different flavours of food and drink are identified according to how much they stimulate these four types of receptor.



**Fig. 1. Sensory nerve endings (receptors) in the skin.** Touch receptors are sensitive to light pressure. They enable us to distinguish between different textures. Pressure receptors are under the dermis and respond to heavy pressure. Pain receptors are branched nerve endings in the epidermis and most other parts of the body. Heat and cold receptors respond to changes in temperature. Hair movements are detected by nerves wrapped around the hair roots.



## 7.3 The nervous system

### Science understanding

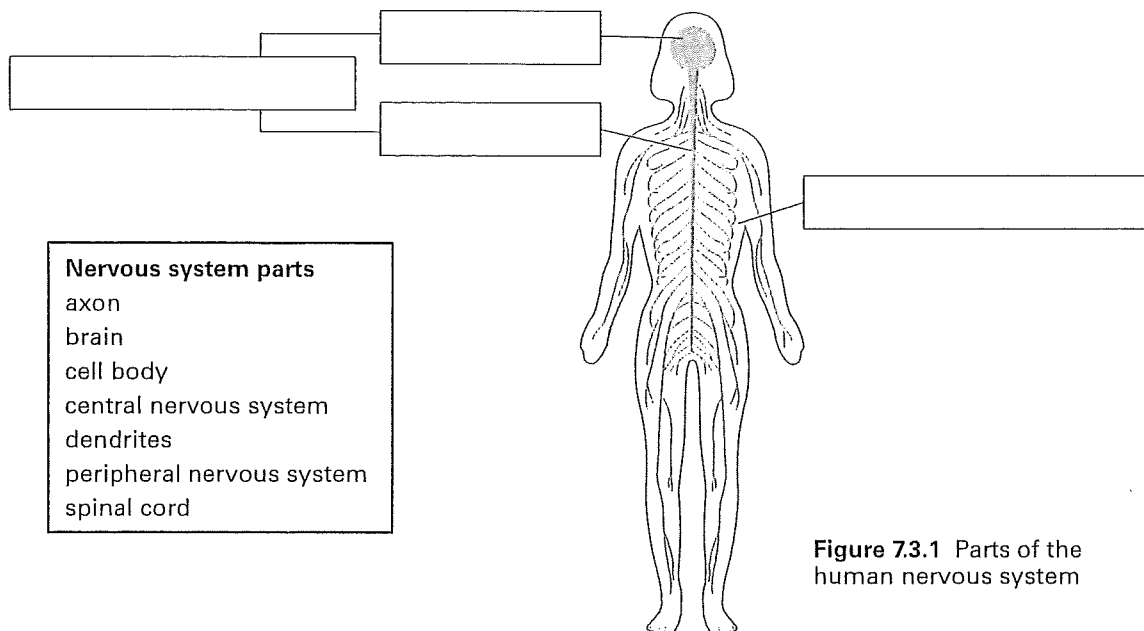
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STANDARD

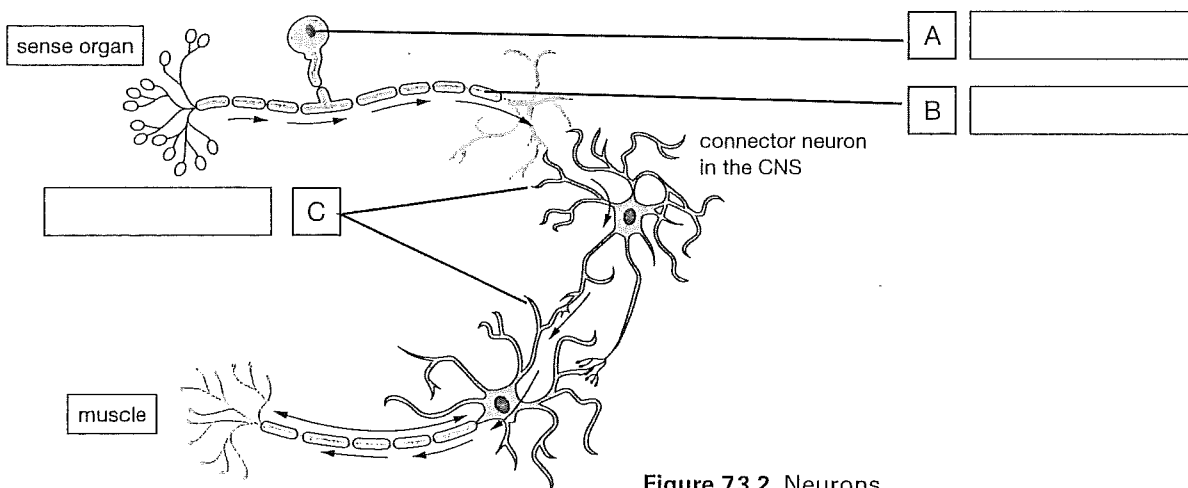
ADVANCED

- 1 Select the correct terms from the box to label the parts of the nervous system (Figure 7.3.1) and the structure of nerves (Figure 7.3.2).

### Parts of the nervous system



### The structure of nerves



- 2 Look at Figure 7.3.2, The structure of nerves, and complete the following labels.
  - (a) Use arrows to show the direction of the nerve impulse.
  - (b) Label the sensory neuron and the motor neuron.
  - (c) Identify and circle two positions where there are synapses.

## DEFINE:

(v) receptor

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(vi) response

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(vii) stimulus

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(viii) thermoreceptor

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## Review Questions

1. Why does the body have receptors?

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2. List some senses for which we have receptors.

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3. Match the following stimuli and receptors.

| Stimulus          | Receptor           |
|-------------------|--------------------|
| touch             | chemoreceptor      |
| taste             | thermoreceptor     |
| stretch in a limb | free nerve endings |
| heat              | proprioceptor      |
| pH of the blood   | osmoreceptor       |

4. The five flavours or stimuli that make up taste are \_\_\_\_\_,  
\_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_ and  
\_\_\_\_\_.

## 7.4 Reflexes

### Science understanding

FOUNDATION

STANDARD

ADVANCED

Reflexes are messages that travel from where they are received, along a sensory neuron to the spinal cord. Within the spinal cord, a relay neuron transmits the message straight back along motor neurons to the first place they were received. This causes a response called a reflex action.

Because reflexes do not have to pass all the way up to the brain and back, they are very fast. Reflexes help protect you from danger. Examples of reflexes are: pulling your hand away from a hot object, blinking your eyes to stop something getting into them, and pulling your foot away after standing on a sharp object. A message is sent to the brain shortly afterwards. Only then can the brain register pain.

- 1 Compare a reflex action with normal stimulus-response reactions of the nervous system.

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- 2 Explain why this makes the reflex action so much faster.

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- 3 Use Figure 7.4.1 to demonstrate the path of a reflex action.

- (a) Identify the pathway of motor neurons in red.
- (b) Identify the relay neuron in green.
- (c) Identify the pathway of the sensory neurons in blue.
- (d) Use arrows to demonstrate the direction of the stimulus.

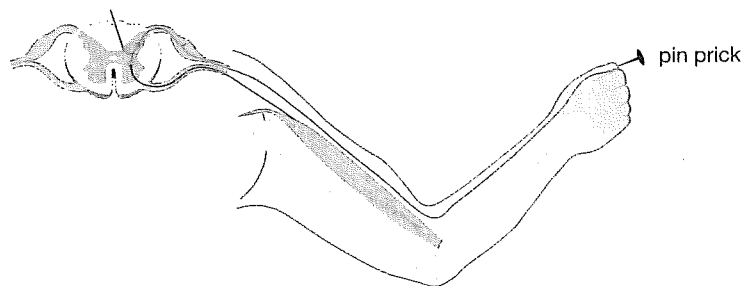


Figure 7.4.1

- 4 Explain why you still feel the pain even after you have moved away from the cause of the pain (stimulus).

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# 7.5 Sweating

## Science understanding

FOUNDATION

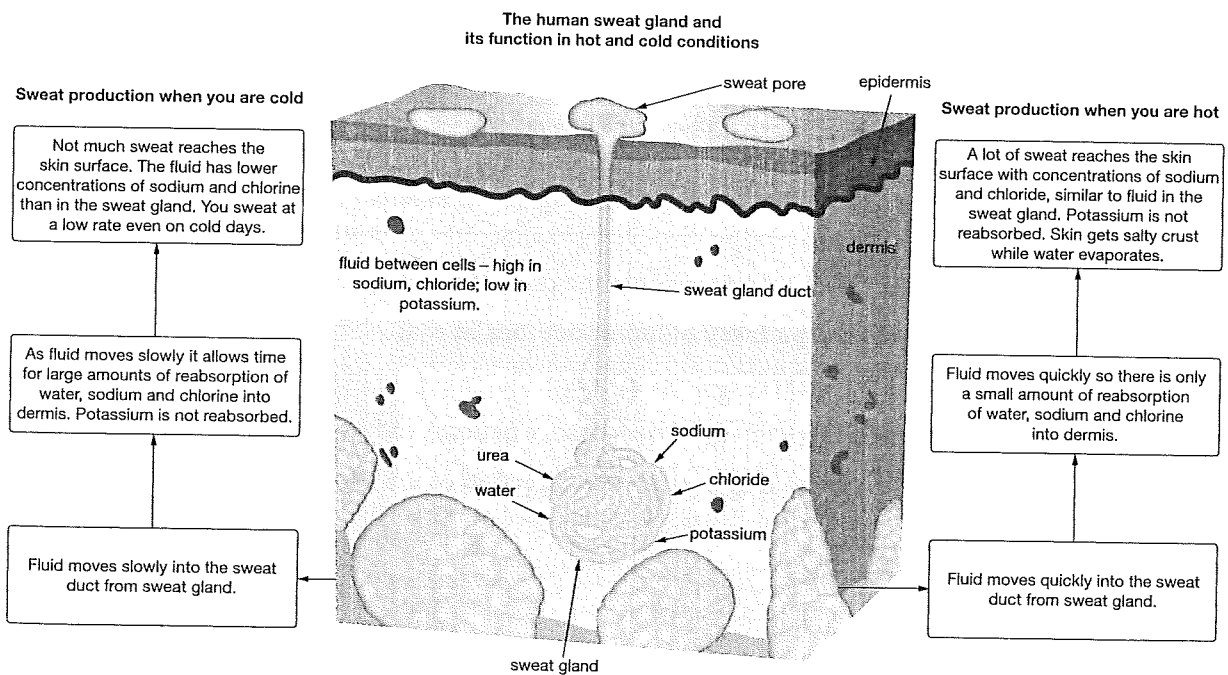
STANDARD

ADVANCED

Your skin and sweat glands are part of the excretory system. Sweat glands are located just under the skin and are found all over your body. Each person has 2 million to 4 million sweat glands. Sweat is essential for survival. Sweating is like a coolant that stops the body from overheating. Excess sweating means your body has lost large amounts of water and salts, resulting in dehydration. The body needs salts to regulate the water content in the blood. It is therefore important to drink water after exercise or when temperatures are very high.

The sweat gland is composed of two parts (Figure 7.5.1). The coiled part is where sweat is produced. The duct leading to a pore on the skin surface is the tube through which sweat is excreted from the body. The fluid that produces sweat comes from the spaces between body cells. This fluid, which is mainly water, has high concentrations of sodium and chloride and low concentrations of potassium and urea. As fluid travels up the sweat duct it changes, depending on conditions such as temperature. The diagram demonstrates these changes when conditions are hot and when they are cold.

**dehydration** (*n*) a lack of water or fluid  
**duct** (*n*) tube or pipe  
**excess** (*adj*) too much  
**regulate** (*v*) to control



**Figure 7.5.1** The human sweat gland

- 1 List the excretory products that leave the body through the skin.

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- 2 Explain where the fluid that forms sweat comes from.

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## Temperature control in the body

The temperature control system of the body is more complex.

- 2 Describe what would happen to the body's feedback system when the core temperature (the temperature within the body) starts to rise.

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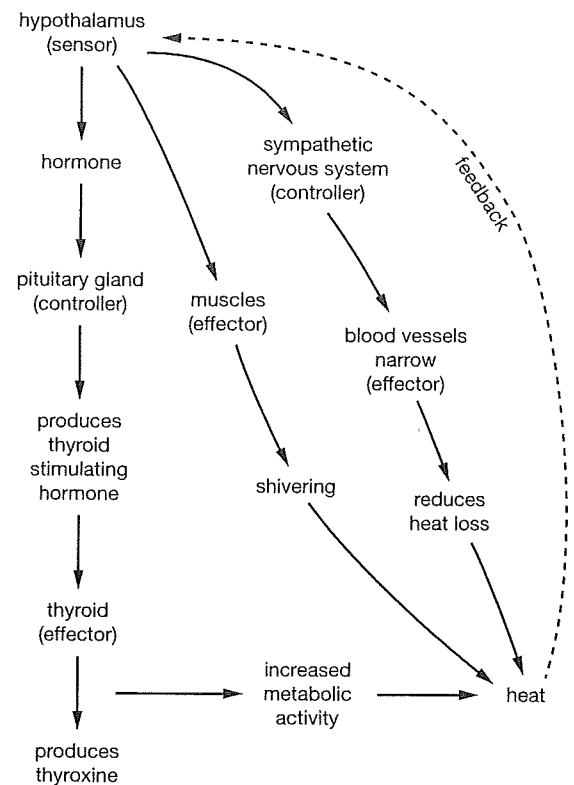
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**Figure 7.9.2** The body's feedback system when the core temperature falls

Your skin also has temperature sensors. These temperature sensors send nerve impulses to the brain. The brain 'tells' the body to react in the same ways as a stimulus from the hypothalamus. The brain also causes the hypothalamus to become more sensitive to changes in core temperature.

- 3 Discuss the advantages of having two types of temperature sensors—the receptors in the skin and the hypothalamus. Next, select the correct phrases from the list and write them in the gaps in the sentences below.

the temperature of your blood is changed  
 there is a change in the external temperature  
 alert your brain to the changes  
 help you maintain a constant temperature  
 behave in ways to help maintain your body temperature

The receptors in the skin react immediately when \_\_\_\_\_.

They \_\_\_\_\_. You can then

\_\_\_\_\_. The hypothalamus detects changes later when

\_\_\_\_\_ rather than just your surface temperature. It will cause

involuntary reactions to \_\_\_\_\_ when you may already be used

to the temperature of the environment.

# 7.10 Literacy review

## Science understanding

FOUNDATION

STANDARD

ADVANCED

Recall your knowledge of human body systems by matching the word in the left-hand column with its correct meaning in the right-hand column.

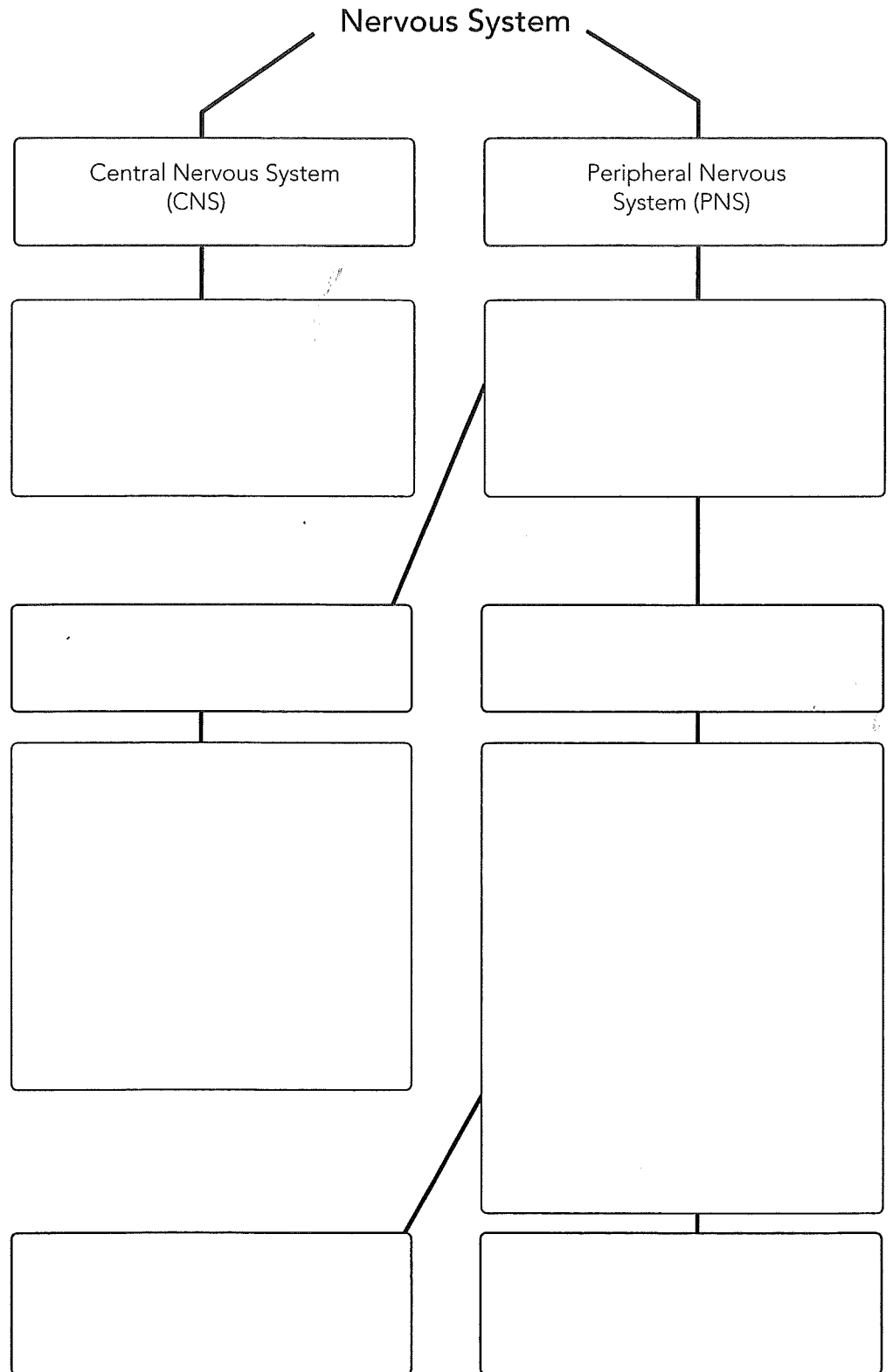
|                           |
|---------------------------|
| autonomic nervous system  |
| brain stem                |
| catalyst                  |
| central nervous system    |
| dendrite                  |
| diffusion                 |
| effector                  |
| endocrine glands          |
| hormones                  |
| hypothalamus              |
| receptors                 |
| mitochondria              |
| neurotransmitter          |
| peripheral nervous system |
| ribosomes                 |
| synapse                   |

|                                                                                                         |
|---------------------------------------------------------------------------------------------------------|
| the brain and spinal cord                                                                               |
| a chemical message released by the axon when the signal reaches the end of a neuron                     |
| a muscle or gland that puts the message into effect                                                     |
| branch from the cell body that receives messages from other neurons                                     |
| a portion of the brain that constantly checks the internal environment of the body                      |
| special cells that detect stimuli                                                                       |
| system controlling involuntary actions such as heartbeat                                                |
| the nerves that carry messages to and from the central nervous system and other parts of the body       |
| movement of particles of a substance from an area of high concentration to an area of low concentration |
| glands that produce hormones                                                                            |
| chemical substances that act as messengers in the body                                                  |
| substance that speeds up the rate of reaction without being used up in the process                      |
| part of the brain that controls vital functions such as breathing                                       |
| the space between neurons                                                                               |
| structures where proteins are manufactured                                                              |
| organelles that are the site of cellular respiration                                                    |

RATE MY UNDERSTANDING  
Shade the face that shows your rating



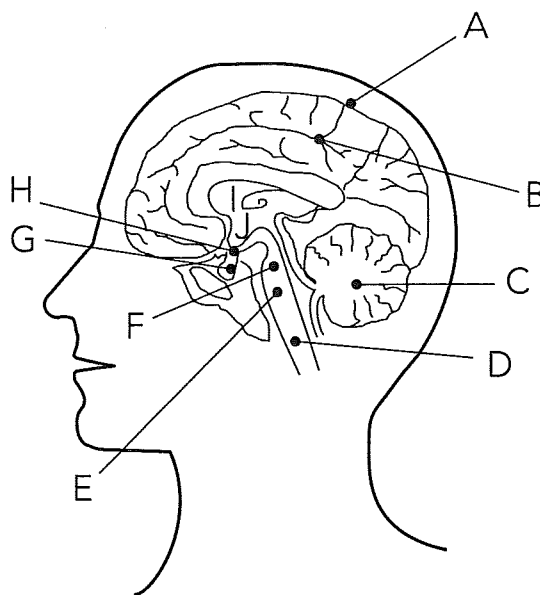
2. Fill in the flow chart below showing the divisions of the nervous system. Beneath each division fill in what it does and what it consists of.





3. Identify the structures shown in the diagram.

- A \_\_\_\_\_
- B \_\_\_\_\_
- C \_\_\_\_\_
- D \_\_\_\_\_
- E \_\_\_\_\_
- F \_\_\_\_\_
- G \_\_\_\_\_
- H \_\_\_\_\_
- I \_\_\_\_\_
- J \_\_\_\_\_



4. What are the functions of these structures?

cerebral cortex (cerebrum)

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cerebellum

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pons (not in the syllabus)

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medulla oblongata

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hypothalamus

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corpus callosum

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